5.12 Exercises

1. Complex arithmetic

Compute $e^{-j\frac{\pi}{2}} + e^{j\frac{\pi}{2}}$.

2. Phase angle

What is the phase angle corresponding to the complex number $1+j$?

3. Discrete convolution

Let $z(t)$ be the convolution of $x(t)$, $0 \le t \le 9$ = 1, 3, 5, 2, 5, 8, 7, 3, 9, 4 and $y(t)$, $0 \le t \le 9$ = 3, 1, 7, 4, 5, 9, 7, 1, 3, 8. Compute $z(5)$.

4. Signals

Give an example of a signal that is continuous, digital, aperiodic, and time-unlimited.

5. Complex exponential

Describe the Im-t projection of the curve $5e^t(\cos 3t + j\sin 3t)$.

6. Linearity

Is the system defined by the transfer function $H(x) = 5\frac{dx}{dt} + 1$ linear?

7. LTI system

What is the output of an LTI system when the input is a real sinusoid?

8. Natural response

Compute the natural response of the LTI system given by
$2\frac{d^2y(t)}{dt^2} + 11\frac{dy(t)}{dt} + 15y(t) = 32x(t)$.

9. Natural response

Determine the exact natural response of the system described by
$2\frac{d^2y(t)}{dt^2} + y(t) = 32x(t); y(0) = 0, \dot{y}(0) = 1$. What is its frequency?

10. Stability

Characterize the stability of the system in Exercise 9.

11. Fourier series

Find the Fourier series corresponding to the infinite series of isosceles triangular pulses of base width 2τ and height 1 spaced apart T_0 seconds. Use the fact that $\int xe^{ax}dx = \frac{axe^{ax} - e^{ax}}{a^2} + C$.

12. Fourier series

Compute the third coefficient of the Fourier series representing a unit periodic rectangular pulse of width $1s$ spaced apart by 10s.

13. Fourier transform

Find the Fourier transform of a single left-triangular pulse defined by $1 - t$ in [0,1].

14. Inverse Fourier transform

Compute the inverse Fourier transform of the function $\pi(\delta(\omega + \omega_0) + \delta(\omega - \omega_0))$ from first principles.

15. Computing the Fourier transform

Use the properties of the Fourier transform to compute the transform of the function $\cos(\omega_0(t + t_0)) + \sin(\omega_0(t - t_0))$.

16. Laplace transform

Compute the Laplace transform of the sinusoid $u(t)\sin\omega_0 t$ from first principles. Locate its pole(s) and zero(es).

17. Laplace transform

Find the Laplace transform of the signal $u(t\text{-}t_0)\sin\omega_0(t - t_0)$.

18. Using the Laplace transform to solve a system

Consider a system whose impulse response $h(t)$ is given by $\cos(\omega_0 t)$. What is its response to the input $e^{-t}u(t)$? Assume that the system is causal.